

LESSON 13 / 第十三課

Supplemental Support Systems: From Crown Load to the Attachment Wound

輔助支撐系統：從樹冠荷載到錨固傷口

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Cables, belts and rods are residual-risk tools. They do not reprint lost section. Static steel limits displacement; dynamic belts allow sway and dissipate energy. Kane and Smiley tested terminations and found the weak link is often the hardware end, not catalogue cable strength. Dee et al. measured changes in modal behaviour, not merely ‘holding the branch up’. Every drilled attachment is a CODIT wound and a fatigue detail. Inspection interval belongs in the specification. Support does not replace a target-based residual-risk sentence.

鋼纜、帶與桿是剩餘風險工具。它們不重印已失截面。靜力鋼限制位移；動力帶容許搖擺並耗能。Kane 與 Smiley 測試錨頭，發現弱點往往是五金端，不是目錄鋼纜強度。Dee 等測得模態改變，而不只是把枝托住。每個鑽孔錨固都是 CODIT 傷口與疲勞細節。檢查間隔屬於規格。支撐不取代基於目標的剩餘風險句子。

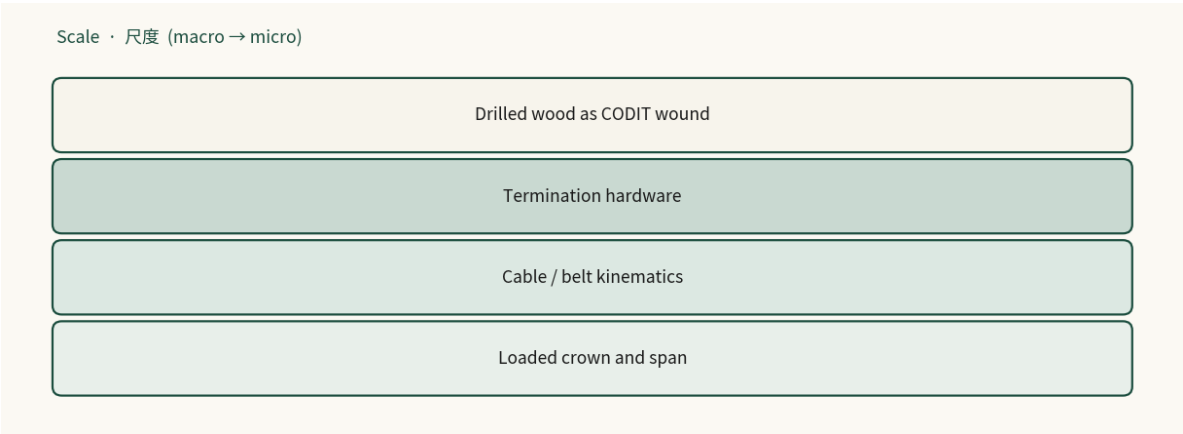


Figure 1. Nested scales from the loaded crown to the attachment wound.

圖 1. 由受載樹冠到錨固傷口的嵌套尺度。

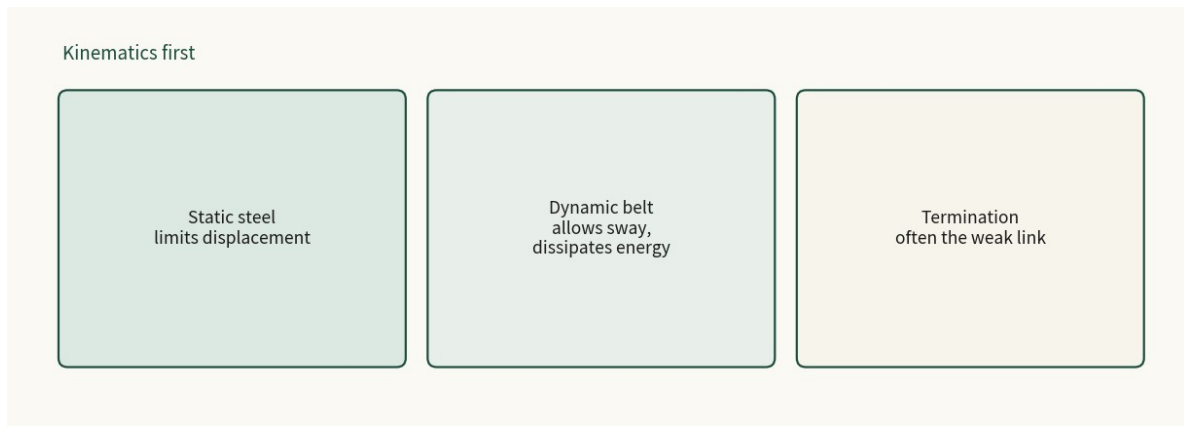


Figure 2. Static, dynamic and termination are different objects.

圖2. 靜力、動力與錨頭是不同對象。

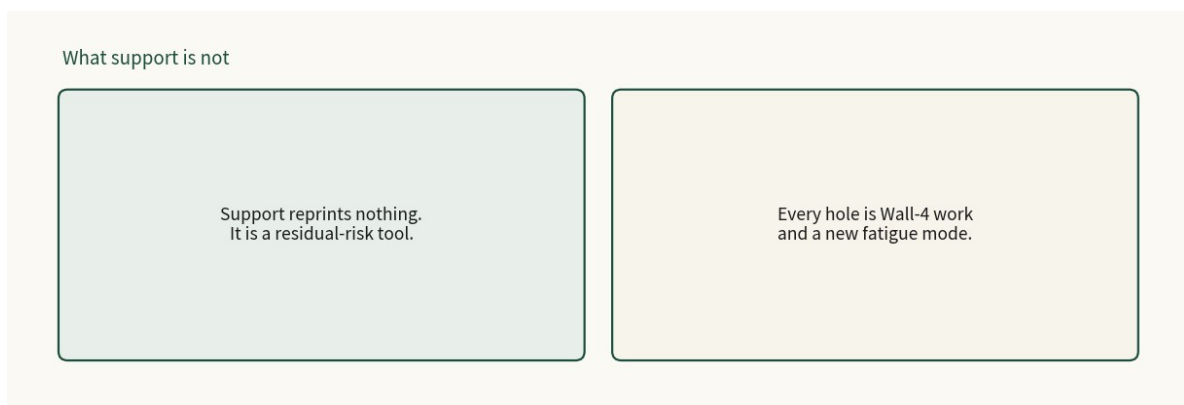


Figure 3. Support is a residual-risk tool, not reprinted section.

圖3. 支撐是剩餘風險工具，不是重印的截面。

1. Scope

1. 範圍

The object is published mechanics of tree support hardware, from the loaded crown to the wood at the fastener.

對象是已發表的樹木支撐五金力學：由受載樹冠到扣件處的木材。

2. System type and termination

2. 系統類型與錨頭

Static steel limits displacement; dynamic belts allow sway and dissipate energy [1,2]. Dee et al. measured changes in modal behaviour under support, which is a different claim from ‘the branch is held up’ [3]. Kane and Smiley’s termination tests located the weak link at the hardware end more often than in catalogue cable strength [4]. Inspection interval and end-of-life of the hardware belong in the specification [1].

靜力鋼限制位移；動力帶容許搖擺並耗能 [1,2]。Dee 等測得支撐下的模態改變，這與「枝被托住」是不同主張 [3]。Kane 與 Smiley 的錨頭測試，把弱點定位在五金端的次數多於目錄鋼纜強度 [4]。檢查間隔與五金壽終屬於規格 [1]。

3. The wound and the sentence

3. 傷口與句子

Drilling is a CODIT event and a stress concentrator. Hardware adds fatigue, girdling and abrasion modes that the unsupported tree did not have. Support changes one term in the risk sentence; it does not abolish targets, weather or residual geometry of the remaining wood.

鑽孔是 CODIT 事件與應力集中。五金增加未支撐之樹所沒有的疲勞、環縊與磨蝕模式。支撐改變風險句子中的一項；它不取消目標、天氣或剩餘木材的剩餘幾何。

4. Limits

4. 限度

Sway series on supported tropical and veteran trees remain short. A catalogue working load is not a measured tree-plus-hardware system capacity in a named storm.

受支撐熱帶樹與老齡樹的搖擺序列仍然短。目錄工作荷載，不是在點名風暴中測得的樹加五金系統容量。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Support systems are:

1. 支撐系統是：

- A. Reprinted section.
A. 重印的截面。
- B. Residual-risk tools.
B. 剩餘風險工具。
- C. P50 raisers.
C. P50 提高器。
- D. Wall 4 itself.
D. 第 4 壁本身。

2. Static vs dynamic first differs in:

2. 靜力對動力首先差在：

- A. Paint.
A. 漆。
- B. Kinematics.
B. 運動學。
- C. Soil pH.
C. 土壤 pH。
- D. 10-20-30.
D. 10-20-30。

3. The weak link is often:

3. 弱點往往是：

- A. Catalogue cable strength.

- A. 目錄鋼纜強度。
- B. The termination.
- B. 錨頭。
- C. Leaf colour.
- C. 葉色。
- D. i-Tree.
- D. i-Tree。

4. A rod hole is:

4. 穿桿孔是：

- A. Free.
- A. 免費。
- B. A CODIT event.
- B. CODIT 事件。
- C. A diversity rule.
- C. 多樣性規則。
- D. P12.
- D. P12。

5. Dee et al. found systems can:

5. Dee 等發現系統可以：

- A. Only hold branches.
- A. 只托枝。
- B. Change sway modes.
- B. 改變搖擺模態。
- C. End inspection.
- C. 結束檢查。
- D. Set occupancy.
- D. 設定佔用。

6. Support replaces the risk sentence:

6. 支撐取代風險句子：

- A. Yes.
- A. 是。
- B. No.
- B. 否。
- C. Only in palms.
- C. 只在棕櫚。
- D. If painted.
- D. 若上漆。

7. Inspection interval belongs in:

7. 檢查間隔屬於：

- A. Folklore.
- A. 傳聞。
- B. The specification.
- B. 規格。

C. P88 tables.

C. P88 表。

D. Canopy percent.

D. 樹冠百分比。

8. Hardware can add:

8. 五金可以增加：

A. Only benefits.

A. 只有益處。

B. Fatigue / girdling modes.

B. 疲勞／環縊模式。

C. Instant Wall 4.

C. 立即第 4 壁。

D. Rain.

D. 雨。

9. A catalogue working load is:

9. 目錄工作荷載是：

A. Measured system capacity in a named storm.

A. 點名風暴中測得的系統容量。

B. Not that measurement.

B. 不是該測量。

C. A tomogram.

C. 斷層圖。

D. A flare datum.

D. 幹基基準。

10. Dynamic belts exist to:

10. 動力帶存在是為了：

A. Abolish sway.

A. 消滅搖擺。

B. Allow sway and dissipate energy.

B. 容許搖擺並耗能。

C. Raise soil oxygen.

C. 提高土壤氧。

D. Set SOS.

D. 設定 SOS。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] ANSI A300 supplemental-support clauses (2023).

[2] Smiley E.T. cabling and bracing BMP / research.

[3] Dee J. et al. (2025). sway and support kinematics.

[4] Kane B., Smiley E.T. termination tests.

[5] James K.R. dynamic properties of trees relevant to support design.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。